

SHREYA SINHA

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Summary

Backend engineer moving into product, with a habit of framing technical decisions as user and business trade-offs. Brings data fluency and close rapport with engineering teams to discovery, prioritisation and outcome-driven delivery.

Education

Master of Computer Applications (MCA)

Birla Institute of Technology, Mesra

CGPA: 8.23

2024 – 2026

Bachelor of Computer Applications (BCA)

Patna Women's College — Platinum Jubilee Scholarship, all six semesters

CGPA: 9.01

2021 – 2024

Technical and Product-Analytical Experience

Network People Services Technologies (NPST)

Jul 2026 – Present

- **Designed authentication and role-based access control** for a 54-feature banking platform with mobile, web, and backend teams; delivered OTP/biometric login, 3 admin roles, and early maker-checker approval, enabling compliance traceability and reusable payment controls.
- **Built a bill-splitting product** in a 2-day assessment to reduce shared-expense disputes; prioritized automated per-member balances and clear error handling, meeting all 6 core requirements.
- **Completed a 6-department cross-functional rotation** and led problem framing for a 22-member team, aligning differing viewpoints into one recommendation.

Research Publications

- **Journal: S. Sinha**, L. Hota, K. K. Senapati, and A. Kumar, “SmartTraffic AI: A Real-Time Adaptive Traffic Signal Control System,” *IEEE Transactions on Networking*, 2026. [**Communicated**]
Tackled urban congestion with a camera-only adaptive signal system, cutting simulated intersection wait time by 38.1% on low-cost hardware.
- **Conference: S. Sinha**, L. Hota, and K. K. Senapati, “SmartTraffic AI for a Real-Time Adaptive Traffic Signal Control System,” *2026 IEEE Future Networks World Forum (FNWF)*, 2026. [**Communicated**]
Improved operator trust in AI signal decisions via plain-language local explanations, reaching 87% confidence in 200 feedback rounds.
- **Conference: S. Sinha**, L. Hota, K. K. Senapati, and P. Paul, “A Real-Time AI-Driven Traffic Management System using Deep Vision and Context-Aware AI,” *4th International Conference on Innovation in Engineering & Management (ICIEM'26)*, BIT Noida, 2026. [**Accepted**]
Turned traffic images into congestion alerts and rerouting advice, choosing YOLOv8 after benchmarking 4 models (96.1% mAP@50).

Projects

Vibeshelf | *Express.js, React, Llama 3.3-70B (Groq), Jaccard Similarity* [Devpost](#)

Jun 2026

- At Mind the Product's **World Product Day hackathon** (2-person team), **led the Express.js backend** for cross-media vibe search, building auth, shelf/list APIs and a feedback-weighted re-ranker tuned against overcorrection, shipping a **live engine matching plain-language moods to 547 books, films and games**.

Fraud Detection System | *Python, FastAPI, Data Pipelines, Anomaly Detection* [GitHub](#)

Jan 2026 – Feb 2026

- With fraud at **2% of ~100K transactions** (a flag-nothing model scores 98%), **benchmarked Isolation Forest, One-Class SVM and an autoencoder** on precision, recall, F1 and AUC-ROC, set the threshold by weighing missed-fraud losses against review load, and **shipped it as a production-ready REST API** with validation and logging.

Smart AI Traffic Management System | *YOLOv8, PyTorch, Flask, LLaMA 3* [GitHub](#)

Feb 2026 – Mar 2026

- After a **Deep Q-Network** (~90% detection accuracy) **cut wait time 30% and raised throughput 25%** vs. a fixed timer, **prioritised operator adoption over further tuning** with a local LLaMA 3 layer explaining every signal decision in plain language, leading to an **ICIEM'26 acceptance**.

Leadership & Volunteering

Volunteer Teacher — School Mentorship Program, Patna Women's College

- Volunteered to close learning gaps at a government all-girls school, **owning instruction for 120+ students** across Classes 7–11 with widely varying learning levels.
- Adapted lesson plans to each grade's level and pace and iterated on delivery using classroom feedback, **improving the student pass rate by 10%**.

Skills

Product & Analysis: Metric Definition, Trade-off Analysis, Success Criteria, Documentation, Stakeholder Communication, SQL

Languages & Data: Python, SQL, JavaScript, Java, C++; PostgreSQL, MySQL, MongoDB

Backend & Systems: FastAPI, Flask, REST APIs, Spring Boot, React, Linux, Git, Docker, Postman

Data & ML: Data Pipelines, PyTorch, Scikit-learn, Anomaly Detection, Computer Vision, Reinforcement Learning, NLP